

A Comparative Evaluation of Encoder and Decoder Language Models with Parameter-Efficient Fine-Tuning (PEFT/LoRA) for Multi-Class Text Emotion Recognition

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Abstract—Text-based emotion recognition presents a fundamental challenge in Natural Language Processing (NLP) due to severe class imbalance, contextual ambiguity, and informal language in short digital texts. Although Encoder-based architectures and recurrent neural networks have traditionally served as baseline solutions, the emergence of generative Large Language Models (LLMs) offers new opportunities to capture complex affective nuances. However, full fine-tuning of these models remains computationally prohibitive. In this work, we propose an empirical evaluation comparing traditional Encoder models (BiLSTM, BERT) against modern intermediate-scale Decoder architectures (Qwen, Gemma, DeepSeek) using Parameter-Efficient Fine-Tuning (PEFT/LoRA). Experiments are conducted on the CARER / Kaggle Emotion dataset, evaluating classification performance through the Macro-F1 metric under controlled hardware constraints.

Index Terms—Text Emotion Recognition, Large Language Models, LoRA, PEFT, Macro-F1, Multi-class Classification.

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TABLE I
TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

^aSample of a Table footnote.



Fig. 1. Example of a figure caption.

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ACKNOWLEDGMENT

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REFERENCES

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